



Tag 14

09.07.2026

Donnerstag

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

GitHub Repo <https://github.com/www42/a5>

AZ-104

Learning Paths:

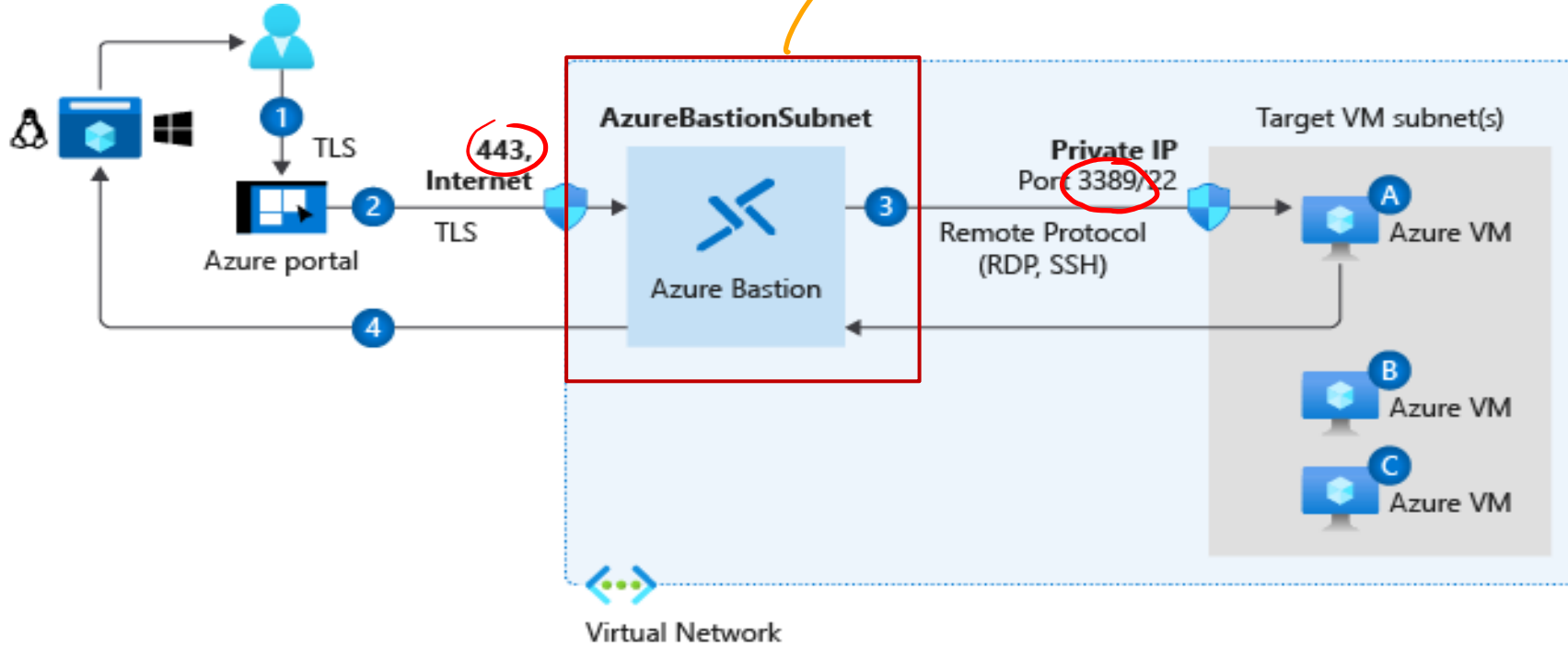
- Manage identities and governance in Azure
- Prerequisites for Azure administrators – CloudShell and templates
- Configure and manage virtual networks for Azure administrators
- Implement and manage storage in Azure
- Deploy and manage Azure compute resources
- Monitor and back up Azure resources

What happened yesterday



Connect to Virtual Machines

Student SKU (Free)
(Developer?)



Bastion Subnet for RDP/SSH through the Portal over SSL

Remote Desktop Protocol for Windows-based Virtual Machines

Secure Shell Protocol for Linux based Virtual Machines

Connect to Windows Virtual Machines

Remote Desktop Protocol (RDP) creates a GUI session and accepts inbound traffic on TCP port 3389

WinRM creates a command-line session so you can run scripts



Connect to Linux Virtual Machines

Administrator account

Authentication type

Username * ⓘ

SSH public key * ⓘ

Provide an RSA public key in the single-line format (starting with "ssh-rsa") or the multi-line PEM format. You can generate SSH keys using ssh-keygen on Linux and OS X, or PuTTYGen on Windows.



[Learn more about creating and using SSH keys in Azure](#)

Authenticate with a SSH public key or password

SSH is an encrypted connection protocol that allows secure logins over unsecured connections

There are public and private keys

Heute



AZ-104

 Deploy and manage
Azure compute resources



Configure Virtual Machine Availability



Setup Availability Sets

Instance details

Name * ⓘ

avset01 ✓

Region * ⓘ

(US) East US ▼

Fault domains ⓘ

2

Update domains ⓘ

5

Use managed disks ⓘ

No (Classic) Yes (Aligned)

Two or more instances in
Availability Sets = 99.95% SLA

Configure multiple
Virtual Machines in
an Availability Set

Configure each
application tier
into separate
Availability Sets

Combine a Load
Balancer with
Availability Sets

Use managed disks
with the Virtual
Machines

Review Availability Zones

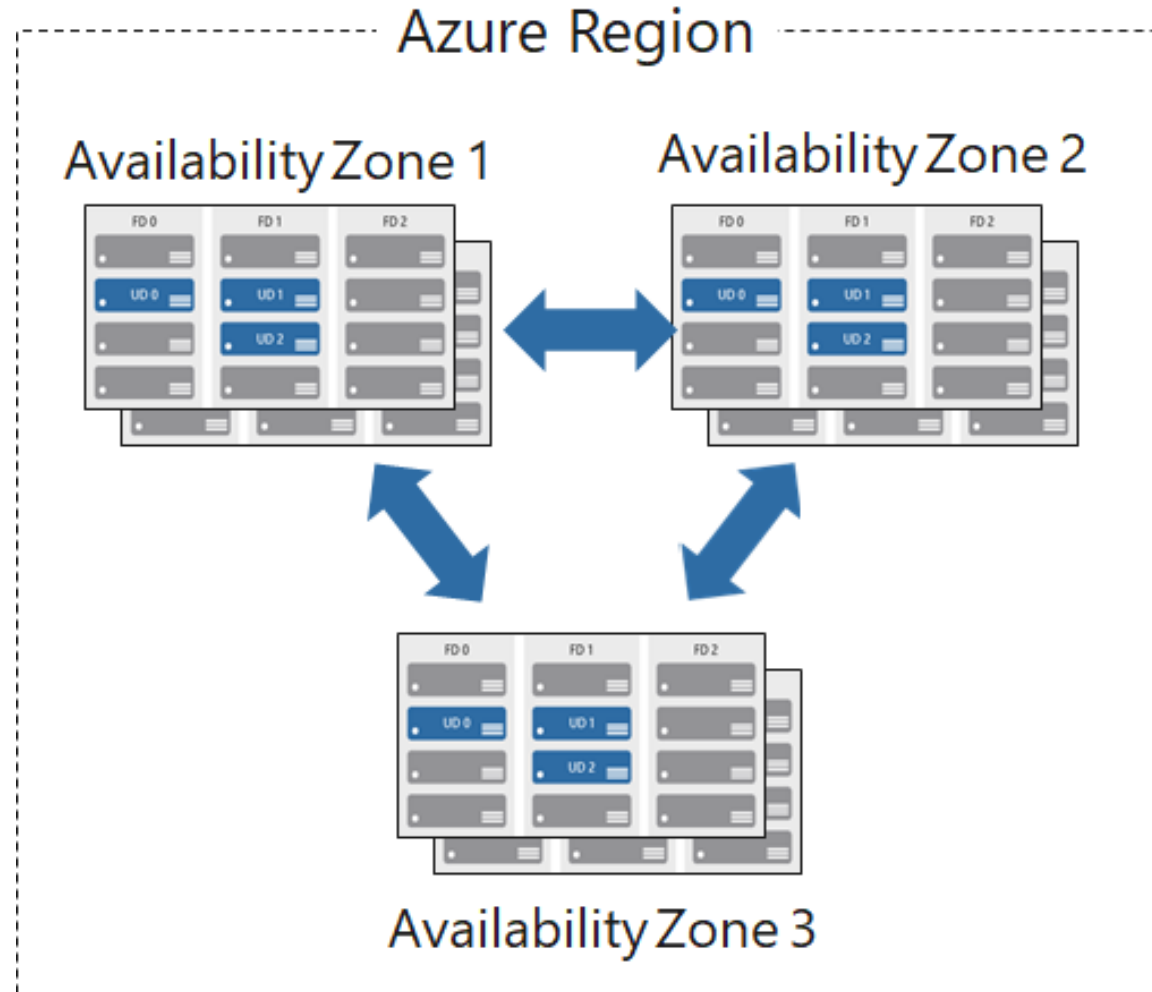
Unique physical locations in a region

Includes datacenters with independent power, cooling, and networking

Protects from datacenter failures

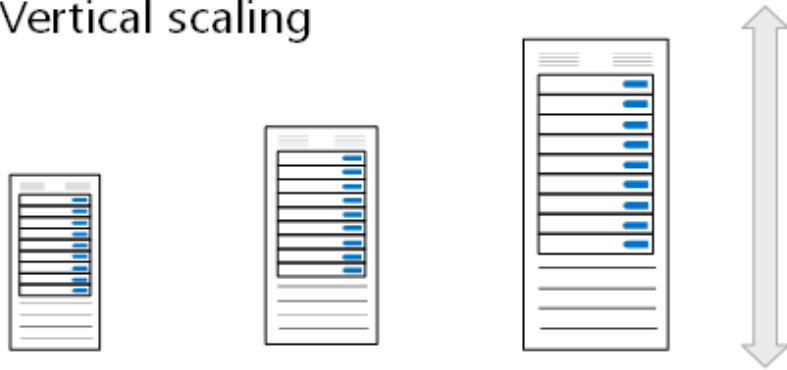
Combines update and fault domains

Provides 99.99% SLA



Compare Vertical to Horizontal Scaling

Vertical scaling



Vertical scaling (scale up and scale down) is the process of increasing or decreasing power to a single instance of a workload; usually manual

Horizontal scaling



Horizontal scaling (scale out and scale in) is the process of increasing or decreasing the number of instances of a workload; frequently automated

Create and configure scaling

- Manually update or autoscale
- Define a minimum, maximum, and default number of VM instances
- Create more advanced scale sets with scale out and scale in parameters

Add a scaling condition

×

Scale mode

☐ Manually update the capacity: Scaling based on a CPU metric, on any schedule

☒ Autoscaling: Scaling based on a CPU metric, on any schedule

Default instance count * ⓘ

Instance limit

Minimum * ⓘ

Maximum * ⓘ

Scale out

CPU threshold greater than * ⓘ

Increase instance count by * ⓘ

Scale in

CPU threshold less than * ⓘ

Decrease instance count by * ⓘ

Query duration

Minutes * ⓘ

The engine will query CPU usage for the past 60 minutes before executing the scaling to avoid reacting to transient spikes.

Schedule

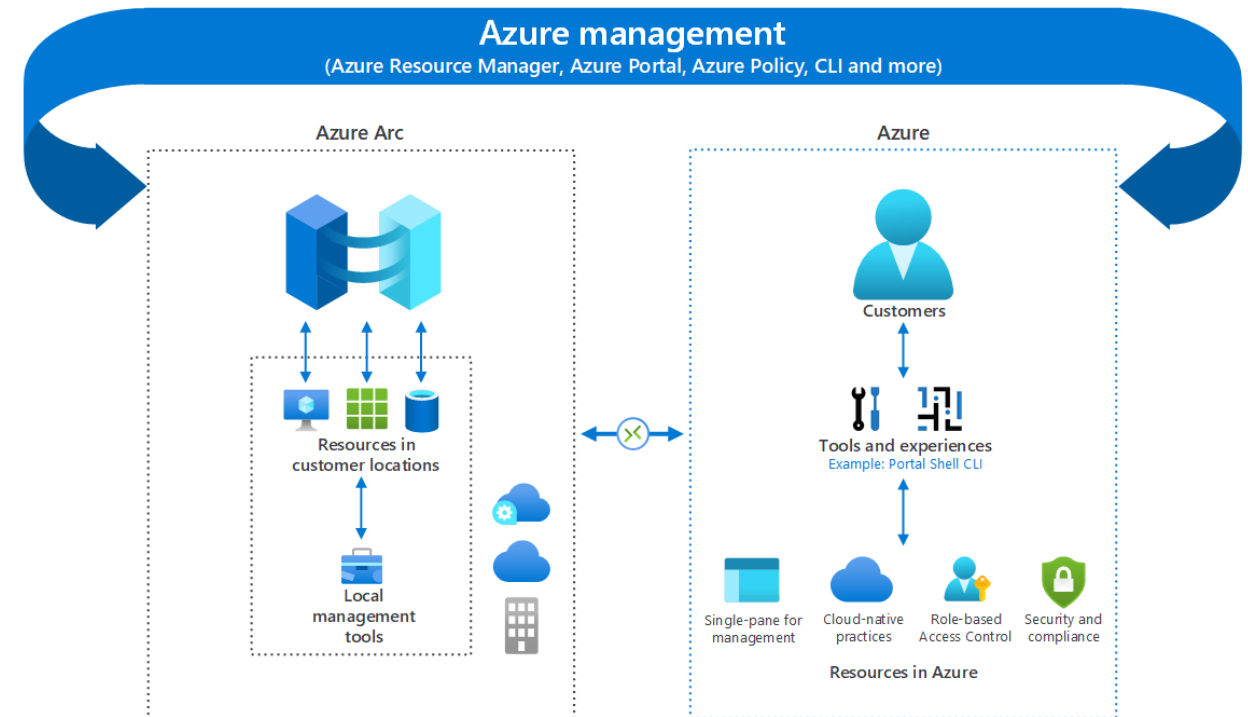
Schedule type *

☒ Specify start/end dates

☐ Repeat specific days

Bring Azure innovation to your hybrid environments with Azure Arc

- Manage your entire environment of existing non-Azure and/or on-premises resources
- Manage virtual machines, Kubernetes clusters, and databases as if they are running in Azure.
- Use familiar Azure services and management capabilities, regardless of where your resources live
- Use traditional IT Ops while introducing DevOps practices to support new cloud native patterns in your environment

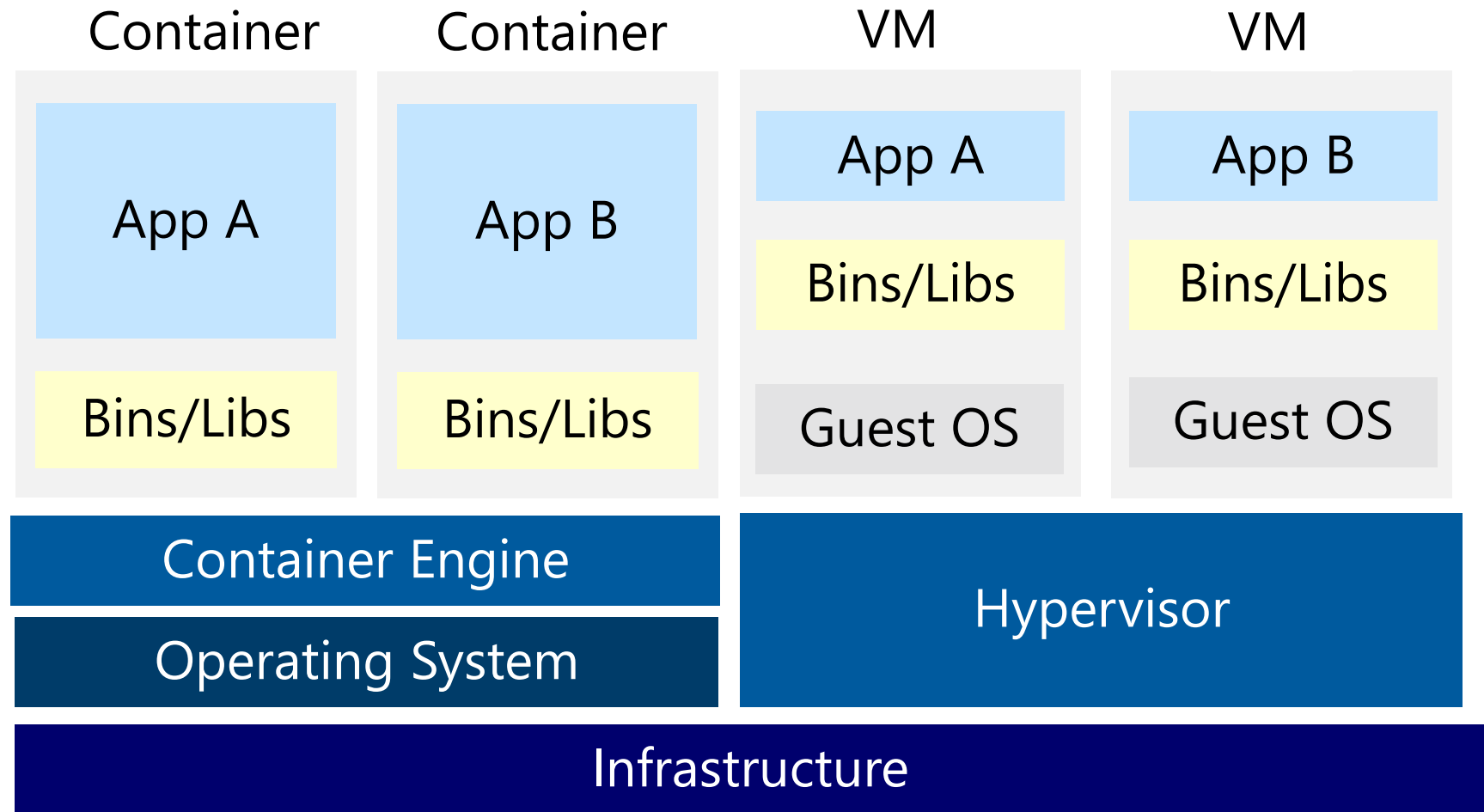


And now for something
completely different...



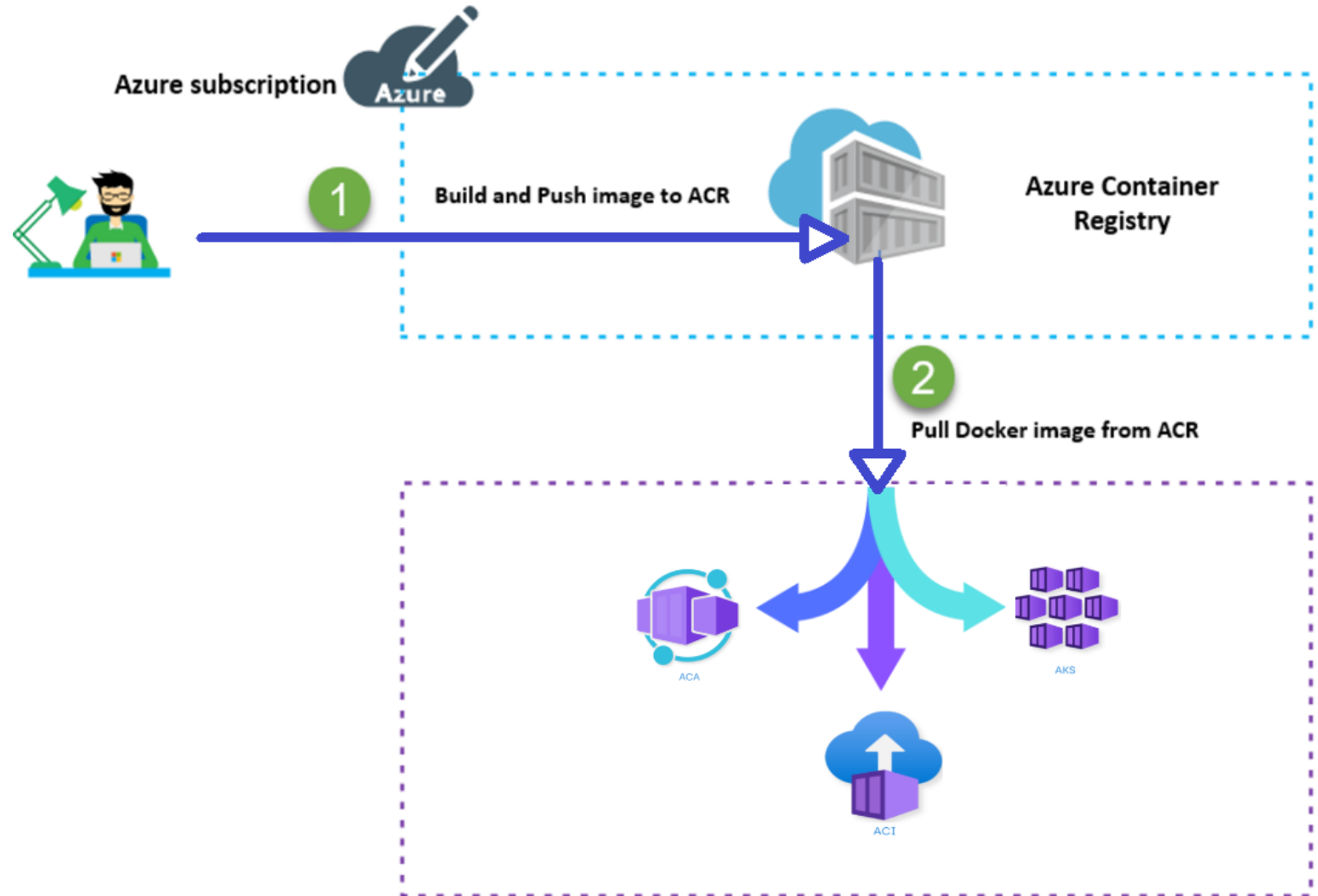
Compare Containers to Virtual Machines

- Isolation
- Operating System
- Deployment
- Persistent storage
- Fault tolerance



Understand Container Images

A container image is a lightweight, standalone, executable package of software that encapsulates everything needed to run an application.



Labs



TO THE LAB ZONE →

NO THROUGH ROUTE
Berlin
Europa
Dublin

SPEEDY ROUTE

4²

lounge

Microsoft
tech-ed
2010

lounge

software diagnostics

E94

SD Studio
for Team Foundation Server
Visualize your code in
a software map.

SD Developer Edit
for C and C++ for Visual
Trace dynamically your code
Visualize the control flow.

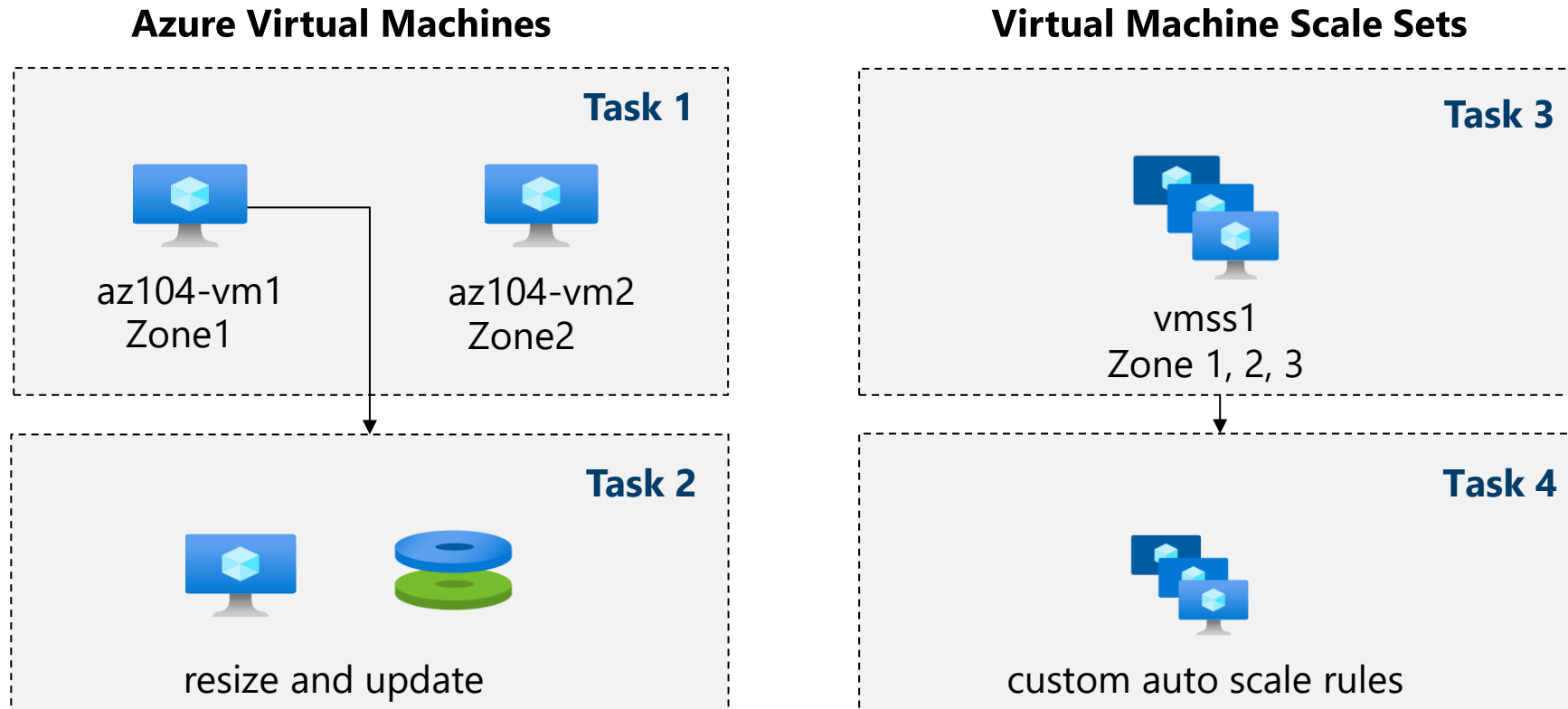
software
diagnostics

DataCore
SOFTWARE

Lab 08 – Manage Virtual Machines



Lab 08 – Architecture diagram



Task 5: Create a virtual machine using Azure PowerShell (option 1)

Task 6: Create a virtual machine using the CLI (option 2)

End of presentation



